Module 5

Exercise 4

1.) Find the customer who ordered the most products (by quantity).

A1.)SQL

AI Assistant Query Query History

```sql
select c.customer_id, c.company_name, sum(od.quantity) as total_quantity
from customers c
join orders o on c.customer_id = o.customer_id
join order_details od on o.order_id = od.order_id
group by c.customer_id, c.company_name
order by total_quantity desc
limit 1;
```

Output

```
 customer_id |     company_name     | total_quantity
-------------+----------------------+----------------
 SAVEA       | Save-a-lot Markets   |           4958
(1 row)
```

2.) List employees and the total number of orders they handled.
A2.)SQL

AI Assistant Query Query History

```sql
select e.employee_id, e.first_name, e.last_name, count(o.order_id) as total_orders
from employees e
left join orders o on e.employee_id = o.employee_id
group by e.employee_id, e.first_name, e.last_name
order by total_orders desc;
```

Output

employee_id	first_name	last_name	total_orders
4	Margaret	Peacock	156
3	Janet	Leverling	127
1	Nancy	Davolio	123
8	Laura	Callahan	104
2	Andrew	Fuller	96
7	Robert	King	72
6	Michael	Suyama	67
9	Anne	Dodsworth	43
5	Steven	Buchanan	42

(9 rows)

3.) Find the most ordered product (by total quantity).

A3.)SQL

AI Assistant Query Query History

```sql
select p.product_id, p.product_name, sum(od.quantity) as total_quantity
from products p
join order_details od on p.product_id = od.product_id
group by p.product_id, p.product_name
order by total_quantity desc
limit 1;
```

Output

```
 product_id |   product_name    | total_quantity
------------+-------------------+----------------
         60 | Camembert Pierrot |           1577
(1 row)
```

4.) Retrieve customers who ordered products from more than one category.
A4.)SQL

AI Assistant Query Query History

```sql
select c.customer_id, c.company_name
from customers c
join orders o on c.customer_id = o.customer_id
join order_details od on o.order_id = od.order_id
join products p on od.product_id = p.product_id
group by c.customer_id, c.company_name
having count(distinct p.category_id) > 1;
```

Output

customer_id	company_name
ALFKI	Alfreds Futterkiste
ANATR	Ana Trujillo Emparedados y helados
ANTON	Antonio Moreno Taquería
AROUT	Around the Horn
BERGS	Berglunds snabbköp
BLAUS	Blauer See Delikatessen
BLONP	Blondesddsl père et fils
BOLID	Bólido Comidas preparadas
BONAP	Bon app'
BOTTM	Bottom-Dollar Markets
BSBEV	B's Beverages
CACTU	Cactus Comidas para llevar
CENTC	Centro comercial Moctezuma
CHOPS	Chop-suey Chinese
COMMI	Comércio Mineiro
CONSH	Consolidated Holdings
DRACD	Drachenblut Delikatessen
DUMON	Du monde entier
EASTC	Eastern Connection
ERNSH	Ernst Handel
FAMIA	Familia Arquibaldo
FOLIG	Folies gourmandes
FOLKO	Folk och fä HB
FRANK	Frankenversand
FRANR	France restauration
FRANS	Franchi S.p.A.
FURIB	Furia Bacalhau e Frutos do Mar

5.) List suppliers whose products appear in orders.

A5.)SQL

AI Assistant Query Query History

```sql
select distinct s.supplier_id, s.company_name
from suppliers s
join products p on s.supplier_id = p.supplier_id
join order_details od on p.product_id = od.product_id;
```

Output

```
 supplier_id |              company_name
-------------+-----------------------------------------
          28 | Gai pâturage
          29 | Forêts d'érables
           5 | Cooperativa de Quesos 'Las Cabras'
          11 | Heli Süßwaren GmbH & Co. KG
          22 | Zaanse Snoepfabriek
           4 | Tokyo Traders
          13 | Nord-Ost-Fisch Handelsgesellschaft mbH
           7 | Pavlova, Ltd.
          25 | Ma Maison
          15 | Norske Meierier
           6 | Mayumi's
          23 | Karkki Oy
          16 | Bigfoot Breweries
           9 | PB Knäckebröd AB
          18 | Aux joyeux ecclésiastiques
          14 | Formaggi Fortini s.r.l.
          21 | Lyngbysild
          10 | Refrescos Americanas LTDA
           8 | Specialty Biscuits, Ltd.
           2 | New Orleans Cajun Delights
          26 | Pasta Buttini s.r.l.
          12 | Plutzer Lebensmittelgroßmärkte AG
          24 | G'day, Mate
           3 | Grandma Kelly's Homestead
           1 | Exotic Liquids
          27 | Escargots Nouveaux
          19 | New England Seafood Cannery
          17 | Svensk Sjöföda AB
          20 | Leka Trading
(29 rows)
```

6.) Find orders that contain products from multiple suppliers.

A6.)SQL

AI Assistant Query Query History

```sql
select o.order_id
from orders o
join order_details od on o.order_id = od.order_id
join products p on od.product_id = p.product_id
group by o.order_id
having count(distinct p.supplier_id) > 1;
```

Output

order_id
10248
10249
10250
10251
10252
10253
10254
10255
10256
10257
10258
10259
10260
10261
10262
10263
10264
10267

7.) Show top 5 customers who placed the most orders.

A7.)SQL

AI Assistant Query Query History

```sql
select c.customer_id, c.company_name, count(o.order_id) as total_orders
from customers c
join orders o on c.customer_id = o.customer_id
group by c.customer_id, c.company_name
order by total_orders desc
limit 5;
```

Output

customer_id	company_name	total_orders
SAVEA	Save-a-lot Markets	31
ERNSH	Ernst Handel	30
QUICK	QUICK-Stop	28
HUNGO	Hungry Owl All-Night Grocers	19
FOLKO	Folk och fä HB	19

(5 rows)